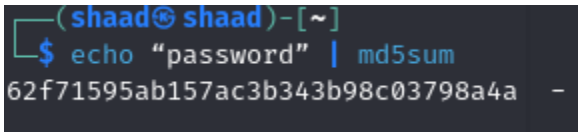
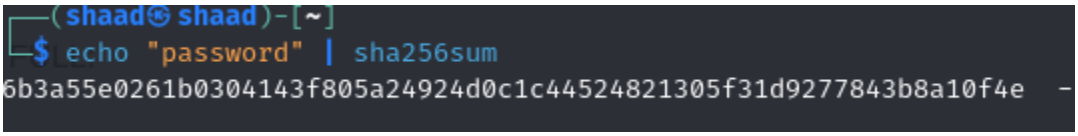
Day – 05 – Cybersecurity Basics

Hashing

echo “password” | md5sum : Generates a MD5 hash of the text "password" and displays its checksum.

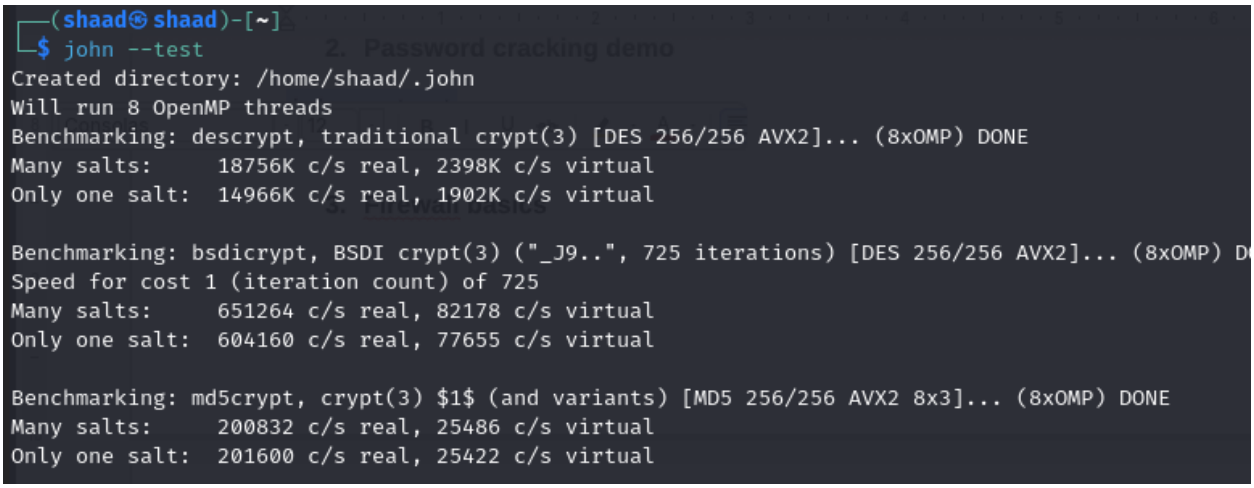


echo "password" | sha256sum : Generates a SHA-256 cryptographic hash of the text "password" and displays its checksum.



Password cracking demo

john --test : Runs benchmark tests in John the Ripper to measure password cracking performance on system



Firewall basics

sudo ufw status : Displays the current status and rules of the UFW

```
┌──(shaad㉿shaad)-[~]
└─$ sudo ufw status
[sudo] password for shaad:
Status: inactive
```

sudo iptables -L : Lists the firewall rules configured in iptables

```
└─$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy DROP)
target     prot opt source               destination
DOCKER-USER  all  --  anywhere             anywhere
DOCKER-FORWARD  all  --  anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

Chain DOCKER (1 references)
target     prot opt source               destination
DROP       all  --  anywhere             anywhere

Chain DOCKER-BRIDGE (1 references)
target     prot opt source               destination
DOCKER     all  --  anywhere             anywhere

Chain DOCKER-CT (1 references)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere             ctstate RELATED,ESTABLISHED

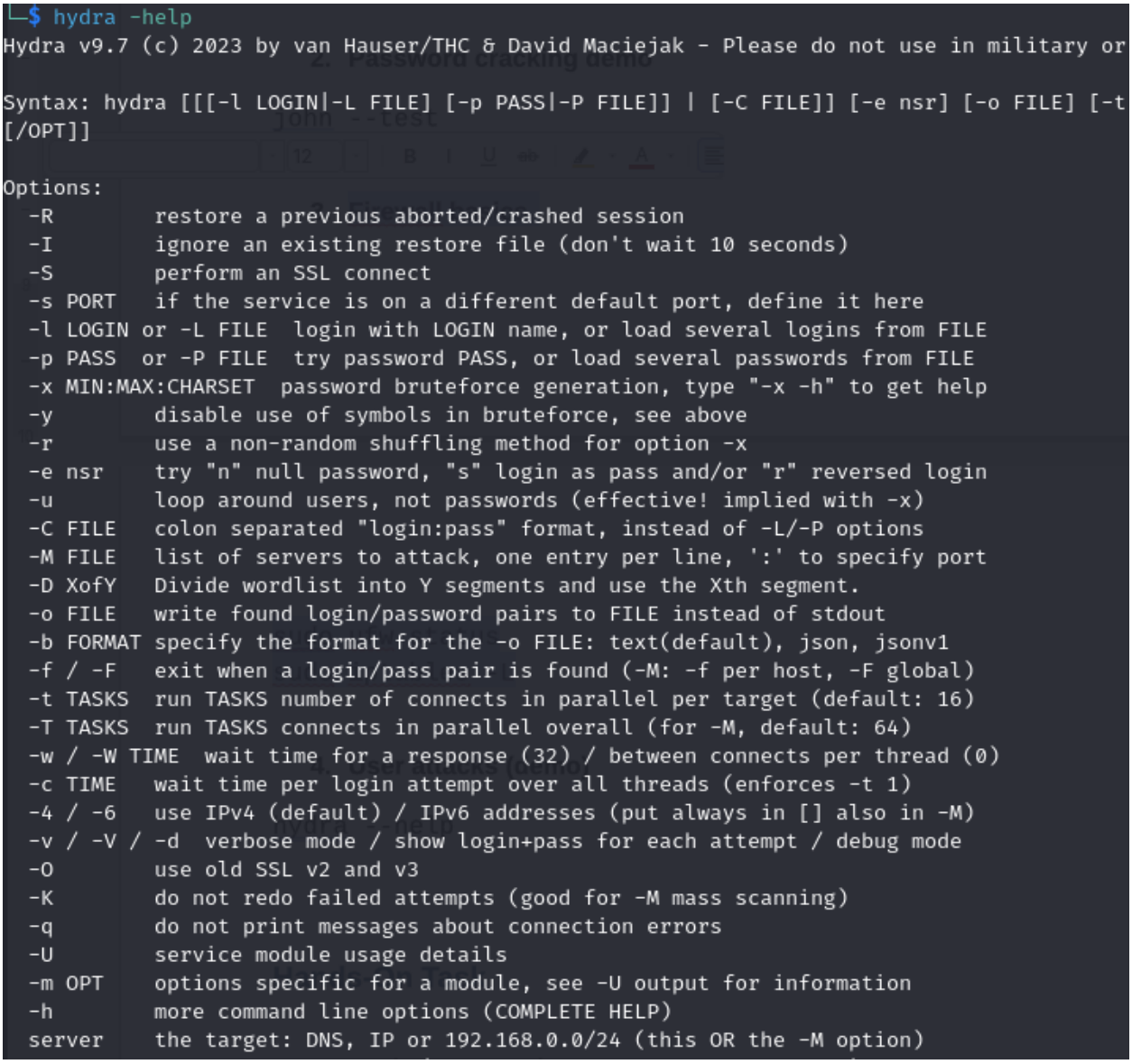
Chain DOCKER-FORWARD (1 references)
target     prot opt source               destination
DOCKER-CT  all  --  anywhere             anywhere
DOCKER-INTERNAL  all  --  anywhere             anywhere
DOCKER-BRIDGE  all  --  anywhere             anywhere
ACCEPT     all  --  anywhere             anywhere

Chain DOCKER-INTERNAL (1 references)
target     prot opt source               destination

Chain DOCKER-USER (1 references)
target     prot opt source               destination
```

User Attacks

Hydra -help : Displays Hydra's help menu, showing available options, supported protocols, and command usage for network login testing.



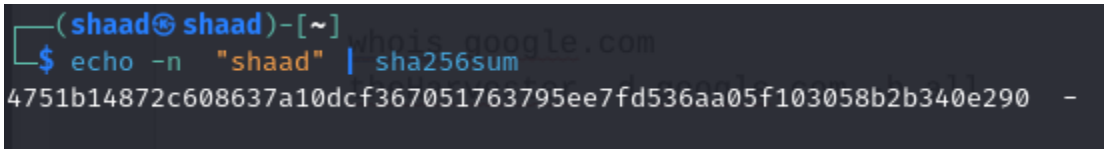
```
└─$ hydra -help
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or

Syntax: hydra [[[-l LOGIN|-L FILE] [-p PASS|-P FILE]] | [-C FILE]] [-e nsr] [-o FILE] [-t
[/OPT]]

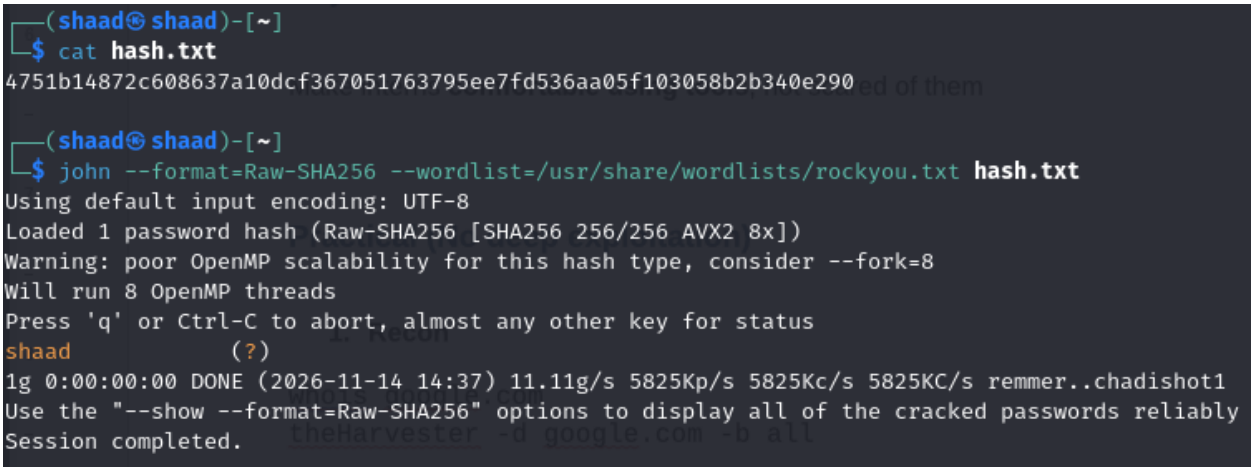
Options:
  -R        restore a previous aborted/crashed session
  -I        ignore an existing restore file (don't wait 10 seconds)
  -S        perform an SSL connect
  -s PORT   if the service is on a different default port, define it here
  -l LOGIN or -L FILE  login with LOGIN name, or load several logins from FILE
  -p PASS  or -P FILE  try password PASS, or load several passwords from FILE
  -x MIN:MAX:CHARSET  password bruteforce generation, type "-x -h" to get help
  -y        disable use of symbols in bruteforce, see above
  -r        use a non-random shuffling method for option -x
  -e nsr    try "n" null password, "s" login as pass and/or "r" reversed login
  -u        loop around users, not passwords (effective! implied with -x)
  -C FILE   colon separated "login:pass" format, instead of -L/-P options
  -M FILE   list of servers to attack, one entry per line, ':' to specify port
  -D XofY   Divide wordlist into Y segments and use the Xth segment.
  -o FILE   write found login/password pairs to FILE instead of stdout
  -b FORMAT specify the format for the -o FILE: text(default), json, jsonv1
  -f / -F   exit when a login/pass pair is found (-M: -f per host, -F global)
  -t TASKS  run TASKS number of connects in parallel per target (default: 16)
  -T TASKS  run TASKS connects in parallel overall (for -M, default: 64)
  -w / -W TIME  wait time for a response (32) / between connects per thread (0)
  -c TIME   wait time per login attempt over all threads (enforces -t 1)
  -4 / -6   use IPv4 (default) / IPv6 addresses (put always in [] also in -M)
  -v / -V / -d  verbose mode / show login+pass for each attempt / debug mode
  -O        use old SSL v2 and v3
  -K        do not redo failed attempts (good for -M mass scanning)
  -q        do not print messages about connection errors
  -U        service module usage details
  -m OPT    options specific for a module, see -U output for information
  -h        more command line options (COMPLETE HELP)
  server    the target: DNS, IP or 192.168.0.0/24 (this OR the -M option)
```

Hands on Task - Day-05

Hash a password : echo -n "shaad" | sha256sum



Crack a weak password (demo wordlist):



Explain difference between hashing & encryption :

Hashing: Converts data into a fixed length hash that cannot practically be reversed to retrieve the original input.

Encryption: Converts readable data into unreadable ciphertext using an encryption key. It can be decrypted using the appropriate key.